

Ray Zirui Zhang

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ACADEMIC POSITIONS	Assistant Professor Department of Mathematical Sciences Worcester Polytechnique Institute	Worcester, USA Aug 2025 - Present
	Visiting Assistant Professor Department of Mathematics University of California, Irvine	Irvine, USA July 2022 - June 2025
RESEARCH SUMMARY	My research interests lie in computational mathematics, scientific machine learning, and data science, with applications in biophysics and cancer research.	
EDUCATION	Ph.D. in Mathematics University of California, San Diego	Sept 2017 – June 2022 San Diego, USA
	M.S. in Computational Science University of California, San Diego	Sept 2017 – June 2019 San Diego, USA
	B.Eng in Civil Engineering University of Hong Kong	Sept 2012 – June 2016 Hong Kong
AWARDS & HONORS	Keynote speaker for the session <i>Computational Models and Methods for Predicting Cancer Progression and Treatment Response</i> at WCCM 2024	2024
	UCI Systems Biology Opportunity Award (with Rany Tith and Song Zhu)	2024
	Rising Star speaker at UCI Center for Complex Biological Systems Retreat	2024
	The Data Science Conference 2023 travel grant (NSF)	2023
	Powell Dissertation Award (UCSD)	2022
	Gold Medal of BirdCLEF 2021 – Birdcall Identification (Kaggle)	2021
	James B. Ax Fellowship (UCSD)	2019
	Emerging Engineering Award (Institute of Civil Engineers)	2017
	Soong Ching Ling Scholarship (Soong Ching Ling Foundation)	2012
PUBLICATIONS & PREPRINTS	BiLO: Bilevel Local Operator Learning for PDE Inverse Problems Ray Zirui Zhang, Christopher Miles, Xiaohui Xie, John Lowengrub. <i>Journal of Computational Physics</i> , 2026	
	Bayesian BiLO: Bilevel Local Operator Learning for Efficient Uncertainty Quantification of Bayesian PDE Inverse Problems with Low-Rank Adaptation Ray Zirui Zhang, Christopher Miles, Xiaohui Xie, John Lowengrub. <i>Submitted. Preprint on arXiv</i>	

Personalized Predictions of Glioblastoma Infiltration: Mathematical Models, Physics-Informed Neural Networks and Multimodal Scans

Ray Zirui Zhang, Ivan Ezhov, Michal Balcerak, Andy Zhu, Benedikt Wiestler, Bjoern Menze, John Lowengrub.
Medical Image Analysis, 2025

Individualizing Glioma Radiotherapy Planning by Optimization of a Data and Physics Informed Discrete Loss

Michal Balcerak, Ivan Ezhov, Petr Karnakov, Sergey Litvinov, Petros Koumoutsakos, Jonas Weidner, Ray Zirui Zhang, John S Lowengrub, Bene Wiestler, Bjoern Menze.
Nature Communications, 2025

A Compact Coupling Interface Method with Second-Order Gradient Approximation for Elliptic Interface Problems

Ray Zirui Zhang, Li-Tien Cheng.
Journal of Scientific Computing, 2024

Binary Level Set method for Variational Implicit Solvation Model

Ray Zirui Zhang, Li-Tien Cheng.
SIAM Journal of Scientific Computing, 2023

Explicit-Solute Implicit-Solvent molecular simulation with Binary level-set, adaptive-mobility, and GPU

Shuang Liu, Ray Zirui Zhang, Li-Tien Cheng, and Bo Li.
Journal of Computational Physics, 2022

Coupling Monte Carlo, Variational Implicit Solvation, and Binary Level-Set for Simulations of Biomolecular Binding

Ray Zirui Zhang, Clarisse G. Ricci, Chao Fan, Li-Tien Cheng, Bo Li, J.Andrew McCammon.
Journal of Chemical Theory and Computation, 2021

Runout Scaling and Deposit Morphology of Rapid Mudflows

Lu Jing, C. Y. Kwok, Y. F. Leung, Ray Zirui Zhang, Lutao Dai.
Journal of Geophysical Research: Earth Surface, 2018

TALKS

Interdisciplinary Center for Quantitative Modeling in Biology, UCR (2025)
The Mathematics of Data, Dynamics, and Life Sciences, UCI (2025)
Center for Computational Mathematics Seminar, UCSD (2025)
Computational and Applied Mathematics Seminar, Arizona State University(2025)
AMS Fall Western Sectional Meeting, UCR (2024)
Computational and Applied Mathematics Colloquium, Penn State (2024)
16th World Congress on Computational Mechanics (WCCM) and 4th Pan American Congress on Computational Mechanics, Vancouver, Canada (2024)

Joint Annual Meeting of the Korean Society for Mathematical Biology and the Society for Mathematical Biology (SMB), Seoul, Korea (2024)
 SIAM Conference on Life Science, Portland, OR (2024)
 2024 UCI Systems Biology Retreat, Pasadena, CA (2024)
 Statistical and Machine Learning Applications in Biomedical Sciences, UCI (2024)
 Center for Computational Mathematics Seminar, UCSD (2024)
 17th U.S. National Congress on Computational Mechanics, Albuquerque, NM (2023)
 The Data Science Conference, University of San Francisco (2023)
 Southern California Applied Mathematics Symposium (2023)
 UCI Applied Math Seminar (2023)
 SIAM Conference on Optimization, Seattle, WA (2023)
 Long Feng Science Forum, The Chinese University of Hong Kong, Shenzhen (2023)
 Predictive Modeling in Biology and Medicine Conference, UC Riverside (2023)
 The International Conference on New Trends in Computational and Data Sciences, Caltech (Poster, 2023)
 SIAM Texas Louisiana Section Annual Meeting, University of Houston (2022)

TEACHING

Instructor, WPI

Aug 2025 – Present

MATH 2631 Probability for Applications (A2025)

Instructor, UCI

Sept 2022 – July 2025

MATH 9 Intro Programming for Numerical Analysis (FA22, SP23, SP24)

MATH 10 Intro Programming for Data Science (SP24, FA24, SP25)

MATH 199 Undergraduate Supervised Reading and Research

Teaching Assistant, UCSD

Sept 2018 – Jun 2022

MATH 114 Intro Computational Stochastics

MATH 170A Intro Numerical Analysis: Linear Algebra

MATH 170B Intro Numerical Analysis: Approximation and Nonlinear Equations

MATH 170C Intro Numerical Analysis: Ordinary Differential Equations

MATH 174 Numerical Methods and Physical Modeling

MATH 180A Intro Probability

MATH 181A Intro Statistics

MATH 270A Numerical Linear Algebra

MATH 270C Numerical Ordinary Differential Equations

SERVICES

Mentor of Math ExpLR: Summer Research Program for High School Students 2024

Organizer of Minisymposium *Image-Guided Mathematical Modeling in Clinical* 2024
and Pre-Clinical Oncology at SMB annual meeting

Guest Lecturer of UCI Cancer Systems Biology Short Course 2024

Reviewer of NeurIPS, ICML, ICLR, TMLR

Organizer of 2023 Southern California Applied Mathematics Symposium 2023